

Clean-Room Evaluation Scale-Up for Unchessed: Leakage-Safe Data and Incremental Fixed-Point Residual Inference

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Abstract—This paper reports the next clean-room stage of the Unchessed evaluation-bar project. The work separates public architectural understanding from implementation: Stockfish 19 is used only as a black-box UCI reference, while all Unchessed features, state updates, data tooling, and residual inference code are original. A deterministic game-level split over real repository PGN archives produced 300,000 legal positions with 239,501 training, 30,984 validation, and 29,515 test records. The nonlinear residual evaluator was refactored from repeated feature extraction to a compact fixed-point context with exact move-delta updates. On the existing seven-position search benchmark, nonlinear NPS improved from the inherited 0.8622 baseline ratio to 0.8962 of HCE NPS, a 3.94 percentage-point recovery, while preserving evaluator correctness. Direct black-box comparison on 120 real-game positions remained 178.925 cp MAE against Stockfish 19 at depth 10. The requested 50/100 parity milestone is therefore not claimed; the evidence supports a measured performance improvement and a data-scale milestone, not SFNNv16 parity.

I. RESEARCH BOUNDARY

The implementation is clean-room. The reference engine is queried through UCI only, and no Stockfish source, neural-network weights, or copied implementation fragments are included in the Unchessed build. The official Stockfish project describes Stockfish 19 as a major release distributed through its download and source channels [1]. The present work uses that release as an external measurement instrument, not as a software dependency.

II. DATA SCALE-UP

The corpus builder assigns a complete game to one split before extracting positions. The assignment key is the relative PGN path and game index, hashed with BLAKE2b and mapped to 80 percent training, 10 percent validation, and 10 percent test. Positions are sampled deterministically every four plies between plies 16 and 180, excluding checks and game-over positions. This prevents near-identical positions from the same game appearing in multiple splits.

The source material comprises high-rated Lichess games, TWIC games, and historical game archives already present in the repository. The manifest stores only FEN and provenance metadata; it contains no teacher scores and no imported engine parameters. This design permits later black-box labeling while keeping data provenance auditable.

TABLE I
LEAKAGE-SAFE REAL-GAME CORPUS MANIFEST.

Split	Games	Positions
Train	15,401	239,501
Validation	2,016	30,984
Test	1,898	29,515
Total	19,315	300,000

III. INCREMENTAL FIXED-POINT RESIDUAL

The previous nonlinear mode recomputed material counts, pawn phase, and king geometry at every evaluation. The new state contains three compact quantities:

$$q = (M, P, K), \quad (1)$$

where M is material imbalance in centipawns, P is total pawn count clipped to 16, and K is Manhattan king distance. The residual is integer-only:

$$R = \text{clip} \left(\frac{I(M, B) + C(B) + G(K, B)(16 - P)}{4}, -120, 120 \right), \quad (2)$$

where B is the base HCE score, I is a bounded interaction term, C is a bounded curvature term, and G is a king-geometry sign and distance term.

For a legal move, material changes are derived from the moving piece, promotion piece, captured piece, or en-passant pawn. Pawn count changes only for captures and promotions. King distance is updated from the two post-move king squares. Castling is handled explicitly, and null moves preserve material and pawn state while refreshing geometry. The implementation therefore avoids a second feature scan and is independent of Stockfish’s accumulator design.

IV. CORRECTNESS AND PERFORMANCE

The Rust core compiled successfully using a temporary Cargo-lock compatibility copy for the available Rust 1.75 toolchain; the repository lockfile was restored unchanged. The evaluator test suite passed, including a new legal-move equivalence test covering every legal start-position move. For every tested move, the delta state exactly matched full recomputation for material, pawn count, and king distance.

The inherited nonlinear NPS ratio was 0.8622 on the same seven-position depth-eight benchmark. The new ratio is 0.8962, recovering approximately 3.94 percentage points of

TABLE II
SEARCH-PATH BENCHMARK AFTER RESIDUAL INCREMENTALIZATION.

Metric	HCE	Nonlinear
Median NPS	1,163,714	1,042,894
Median time (ms)	37	37
NPS ratio	1.0000	0.8962

TABLE III
FIXED-CONTROL SMOKE RESULTS FOR THE OPTIMIZED NONLINEAR MODE.

Movetime	Games	Score	Result
30 ms	1	0.500	Draw
100 ms	1	0.500	Draw
300 ms	1	0.500	Draw

the prior 13.78 percentage-point overhead. This is a meaningful hot-path improvement, but it does not yet eliminate the bottleneck. The remaining cost is dominated by the base HCE evaluation and search-mode integration rather than the three residual scalars alone.

V. BLACK-BOX STRENGTH AND SCORE VALIDATION

A direct UCI-only comparison was rerun against the official Stockfish 19 Linux x86-64 universal binary. On the existing 120 legal real-game positions at depth 10, the homemade mode measured 178.925 cp mean absolute error, 103 cp median absolute error, and 1,204 cp maximum absolute error. These values are consistent with the inherited score-parity measurement and show that the optimization preserved behavior rather than changing the model’s calibration.

A fixed-control smoke tournament used one game per variant at 30, 100, and 300 ms per move, with an 80-ply cap. The nonlinear mode drew all three sampled games against Stockfish 19. The sample is far too small for an Elo claim, but it provides a protocol-level regression check: the optimized mode produced legal play and did not exhibit an immediate stability failure.

VI. PARITY ASSESSMENT AND LIMITATIONS

The project should not claim SFNNv16 parity from these results. The 300,000-position corpus is a data-scale milestone, not a trained neural model. The residual remains a handcrafted nonlinear correction over the HCE backbone, and the 120-position score corpus remains the primary direct calibration set. The present evidence supports a practical parity estimate of approximately 23/100, up from the inherited 20/100 assessment, reflecting improved inference integration and reproducibility rather than a demonstrated strength leap. The requested 50/100 milestone remains open.

The next technically justified step is black-box teacher labeling of the leakage-safe train and validation splits, followed by a compact original model with integer feature transforms, per-layer scales, and calibration against held-out games. Any parity claim should then report held-out MAE, rank correlation,

calibration error, NPS, and a sufficiently powered fixed-control match rather than relying on one aggregate score.

VII. CONCLUSION

The scale-up is successful in two narrow, measurable senses. First, the clean-room research facility now has a 300,000-position, game-level leakage-safe manifest sourced from real games. Second, the nonlinear search integration now uses exact move-delta updates and improves its NPS ratio from 0.8622 to 0.8962 on the established benchmark. Direct Stockfish 19 score parity remains approximately 178.925 cp MAE, so SFNNv16-quality evaluation has not been reached. The work is therefore a validated engineering advance and an honest foundation for the next teacher-labeled, quantized model experiment.

REFERENCES

- [1] Official Stockfish, “Stockfish 19,” Stockfish official download and release information, 2026. [Online]. Available: <https://stockfishchess.org/download/>
- [2] G. Wu, “NNUE: Efficiently Updatable Neural Network,” Chessprogramming Wiki, accessed 2026. [Online]. Available: <https://www.chessprogramming.org/NNUE>